```
View(diamonds)
# Import Placement_Data_Full_Class
str(Placement_Data_Full_Class)
summary(Placement_Data_Full_Class)
is.na(Placement_Data_Full_Class)
sum(is.na(Placement_Data_Full_Class))
which((is.na(Placement_Data_Full_Class)))

#Central Tendencies
Mean=mean(Placement_Data_Full_Class$salary, na.rm = TRUE)
Median=median(Placement_Data_Full_Class$salary,na.rm = TRUE)

install.packages("modeest")
library(modeest)
Mode= mfv(Placement_Data_Full_Class$salary, na_rm = TRUE)
print(Mode)

SD=sd(Placement_Data_Full_Class$salary, na.rm = TRUE)
Variance= var(Placement_Data_Full_Class$salary, na.rm = TRUE)

Quartile= quantile(Placement_Data_Full_Class$salary, na.rm = TRUE)
print(Quartile)
Range= IQR(Placement_Data_Full_Class$salary,na.rm = TRUE)
print(Range)
#Coefficient of variation
sd(Placement_Data_Full_Class$salary,
na.rm=TRUE)/mean(Placement_Data_Full_Class$salary, na.rm=TRUE)*100

library(pastecs)
stat.desc(Placement_Data_Full_Class)
#It displays all values after omiting NA
a=c(2,4,6,10)
b=c(12,25,30,50)
sd1=sd(a)
sd2=sd(b)
cv1=sd1/mean(a)*100
cv2=sd2/mean(b)*100
# cv1=62.102, cv2=53.9387,
View(diamonds)
summary(diamonds)
stat.desc(diamonds)
trial=data.frame(A=c(10,14,18,20,25),B=c(9,15,18,10,30))
```

```
stat.desc(trial)#CV is more for B so A is more stable.

#Visualize dispersion of price according to cut of the diamonds

ggplot(diamonds, aes(price))+geom_histogram(fill="green")+
  geom_vline(aes(xintercept=mean(price), color="red"))+
  facet_wrap(~cut)

ggplot(diamonds, aes(price))+geom_boxplot(fill="green")+
  geom_vline(aes(xintercept=mean(price), color="red"))+
  facet_wrap(~cut)

ggplot(diamonds, aes(price))+geom_histogram(fill="blue")+
  geom_vline(aes(xintercept=median(price), color="red"))+
  facet_wrap(~cut)
View(diamonds)

A=diamonds %>% group_by(cut) %>% summarise(Mean=mean(price),
Meadian=median(price), SD=sd(price), Variance=var(price))

#to compute the standard deviation (or variance) of multiple
variables at the same time
#lapply: Apply a function over list or a vector
D1=select(diamonds, -cut, -color,-clarity)
lapply(D1[, 1:4], sd)
lapply(D1, sd)#for all columns
lapply(D1[, 1:4], mean)
lapply(D1[, 1:4], var)
lapply(D1[, 1:4], median)

#Skewness
install.packages("moments")
library(moments)
skewness(D1)
skewness(D1$price) #positively skewed
ggplot(D1, aes(price))+geom_density() #right skewed

skewness(D1$carat)  #positively skewed
ggplot(D1, aes(carat))+geom_density() #right skewed

skewness(iris$Petal.Width) #negatively skewed
ggplot(iris, aes(Petal.Width))+geom_density()
```

```
#Kurtosis
kurtosis(D1)
kurtosis(D1$price)  #Leptokurtic
kurtosis(D1$carat)  #Leptokurtic
kurtosis(iris$Petal.Width) #Platykurtic

#Correlation
Data1=select(iris,-Species)
cor(Data1)
D2=cor(Data1)
install.packages("corrplot")
library(corrplot)
corrplot(D2)
corrplot(D2, method="number")
corrplot(D2, method="pie")
```